Chapter 7

Trajectory Generation

(Slides courtesy: Dr. F. Firmani)

Overview

In this chapter the following topics will be covered:

- Types of Trajectory Generation
- Definition of Trajectory Generation
- Types of Motion: Point to Point and Continuous Path
- Point to Point
  - Cubic, Quintic and Trapezoidal Schemes
- Continuous Path
  - Via Point and Cartesian

Types of Path Generation

The tasks of industrial robots can be taught in three different ways:

1. Online programming
2. Off-line programming
3. Sensor-based programming

Online Programming:

The task of the robot is programmed online using a teach pendant. Several points (configurations of the manipulator) are defined and saved. Then the robot is asked to move sequentially through all the points.

(Directindustry.com, 2016)

Types of Path Generation

Off-line Programming:
Software is used to generate the codes that provide the trajectory of the task. Displacements are input as codes. Each manufacturer of robot hardware uses their own software (see Table).

Other Languages such as C, Java, Python, are also commonly used.

Robot brand	Language name
ABB	RAPID
Comau	PDL2
Fanuc	Karel
Kawasaki	AS
Kuka	KRL
Stäubli	VAL3
Yaskawa	Inform

(Wikipedia, 2016)

Types of Path Generation

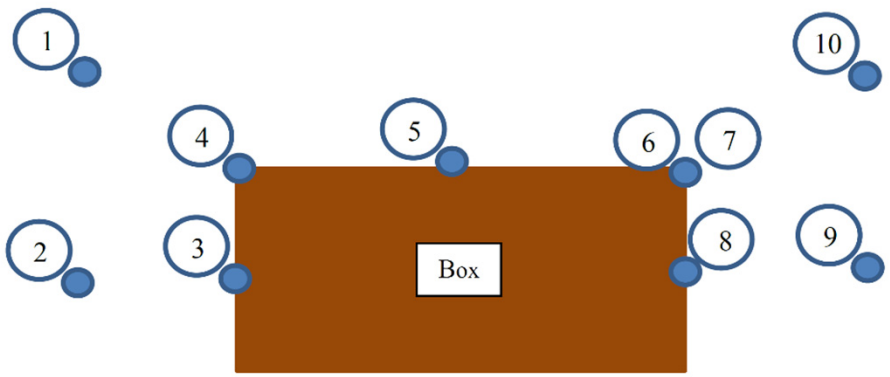
Sensor-based Programming:
This can be accomplished by hand guiding the manipulator through space and the sensors record each position. The robot will eventually replicate the motion.

Autonomous robots accomplish tasks in hostile environments by making decisions depending on the readings captured by sensors.

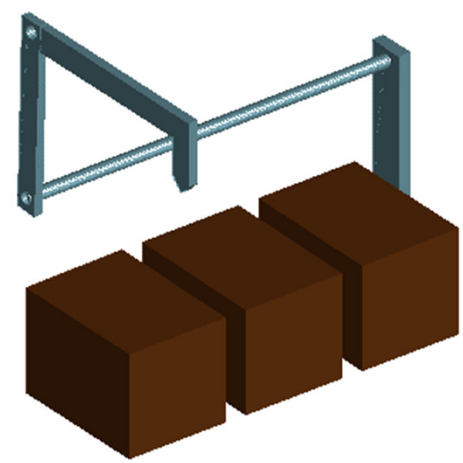
(Robotiq.com, 2016)

Path Generation

In the MSE 429's project, the configurations will be established by defining the desired path of the end-effector.



Point #	Description	X [mm]	Y [mm]	Z [mm]
1	Top	600	-520	920
2	Approach	125	-540	920
3	Box1	125	-500	920
4		250	-500	920
5		250	-325	920
6		250	-150	920
7		250	-150	920
8		125	-150	920
9	Return	125	-135	920
10	Up	300	-135	920

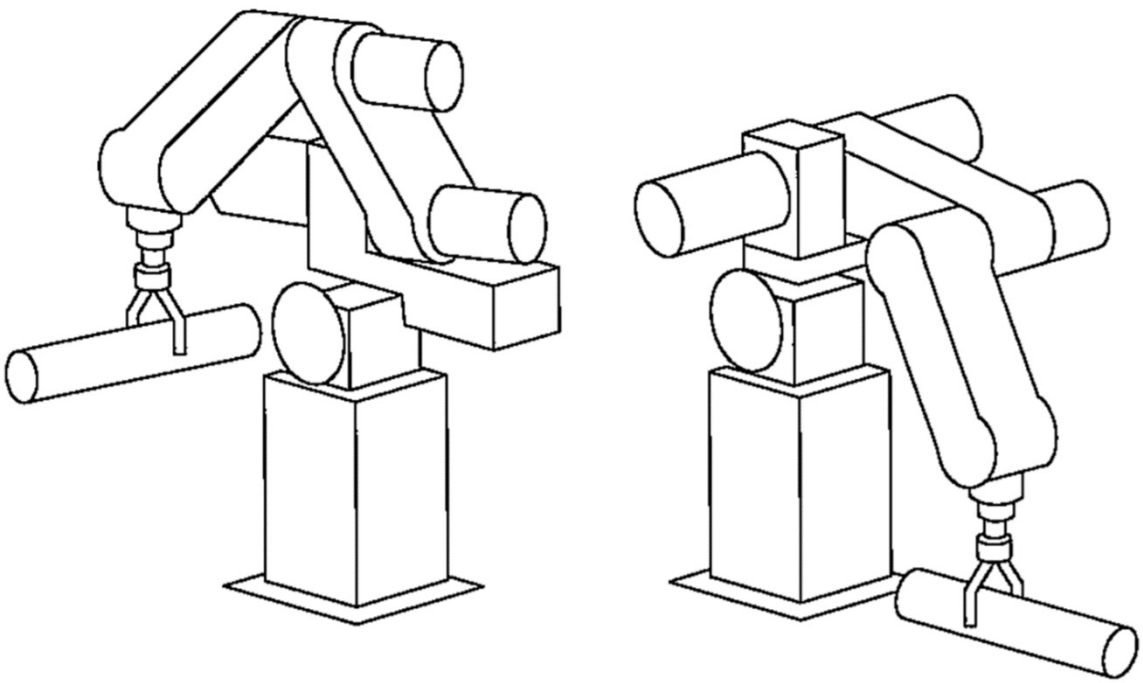


Courtesy of Andy Huerlimann and Alex Vargas, Box Taper 2012

Trajectory Generation

The basic problem of trajectory generation is how to move the manipulator from an initial to a final position and orientation.

A **trajectory** describes the desired motion of the manipulator. This is position, velocity and acceleration of each degree of freedom.



(Trajectory Generation, Craig 2003)

Mathematical Schemes

In order to establish the time history of a change in the position and orientation of the manipulator, mathematical functions or schemes are employed. Common schemes are polynomials.

By taking the time derivative of the mathematical schemes, also the time history of the velocity and acceleration can be determined.

$$\theta(t) = f(t) \qquad \dot{\theta}(t) = f'(t) \qquad \ddot{\theta}(t) = f''(t)$$

In order to have a smooth motion, the second time derivative of the scheme function (acceleration) must be continuous, rough jerky motions tend to wear out the device.

This approach is applied for each joint individually, say joint k:

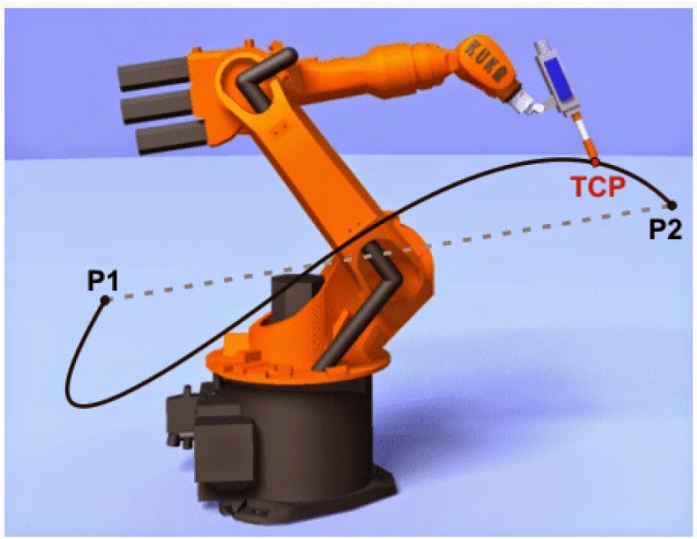
$$\theta_k(t) = f(t) \qquad \dot{\theta}_k(t) = f'(t) \qquad \ddot{\theta}_k(t) = f''(t)$$

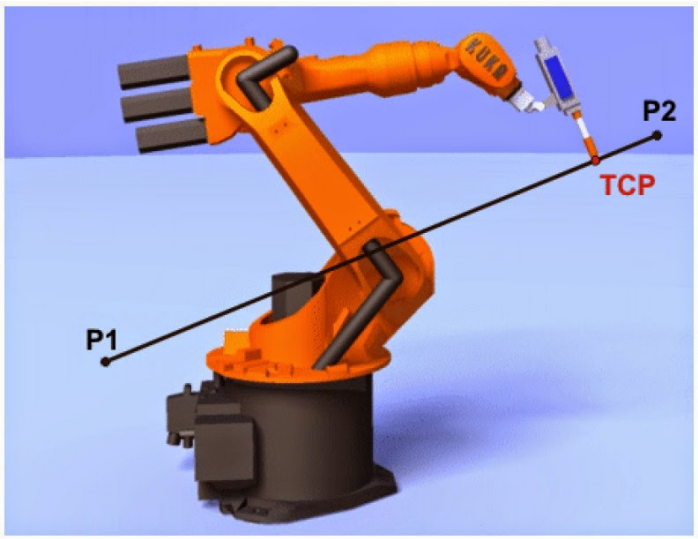
Types of Motion

There are two types of motion between two desired configurations:

Point-to-point (PTP): This trajectory is based on the joint variables. Each joint moves independently from each other, therefore the resulting end-effector path is not controlled.

Continuous path (CP): The goal is to move the end-effector linearly along a particular path.





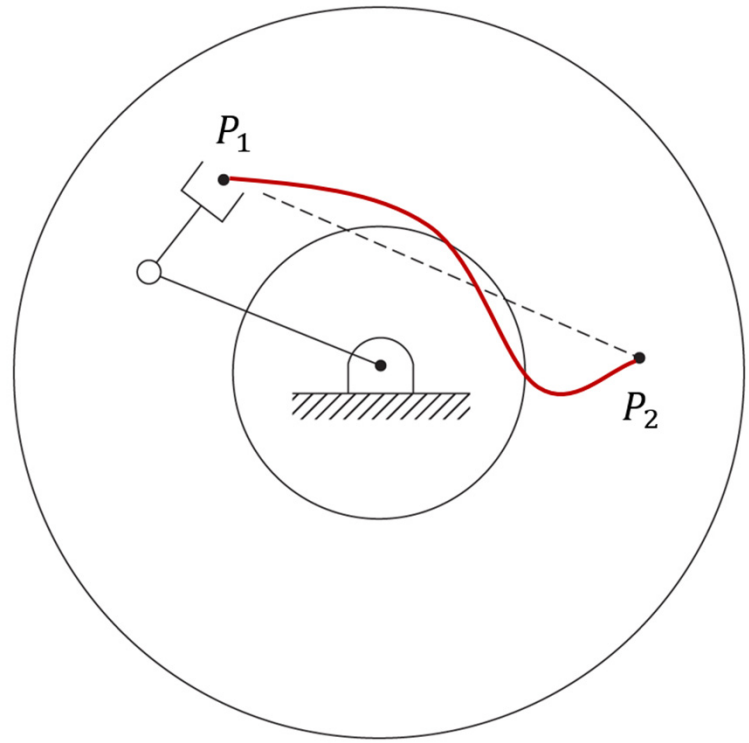
(Kuka's Motion Types, the tool centre point (TCP) can connect two points moving PTP or CP)

Point-to-point Motion

In point-to-point (PTP) motion, each joint moves independently from each other, consequently the path followed is a function of the joint displacements.

Since the desired path is usually established in terms of the end-effector space variables, the inverse kinematics is applied to describe the path in terms of the joint variables.

The time required to move from point P_1 to point P_2 must be the same for each joint. For example, for a 2 revolute jointed robot, let $P_1 = [30 \quad 45]^\top$ and $P_2 = [70 \quad 25]^\top$, thus the change of displacement between postures is $\Delta P = [40 \quad -20]^\top$ implying that joint 1 has to move twice as fast as joint 2.



Cubic Polynomial Scheme (PTP)

This is the simplest mathematical scheme. The goal is to establish the history of change in displacement, velocity and acceleration of a joint during a specific period of type using a cubic polynomial of the form:

$$q_k(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \text{ (Eq. 1)}$$

where, q is the type of joint (θ for revolute and d for prismatic), k is the joint number, a_i are coefficients (to be determined) and t varies from an initial time t_0 to a final time t_f. If a_i are known the position of the joint at any time between t_0 and t_f can be determined.

The velocity and acceleration of the joint are given by

$$\dot{q}_k(t) = a_1 + 2a_2 t + 3a_3 t^2 \text{ (Eq. 2)}$$

$$\ddot{q}_k(t) = 2a_2 + 6a_3 t \text{ (Eq. 3)}$$

Cubic Polynomial Scheme (PTP) – *cont'd*

In order to evaluate coefficients, we require as many constraints as needed. Thus, for a cubic polynomial with four coefficients, four constraints can be satisfied.

Two constraints define the initial and final positions of the joint, i.e.

$$q_k(t_0) = q_{k_0} \qquad \text{(Eq. 4)}$$

$$q_k(t_f) = q_{k_f} \qquad \text{(Eq. 5)}$$

The other two constraints can be used to specify the velocity of the joints, for example assume that the initial and final velocity are zero.

$$\dot{q}_k(t_0) = 0 \qquad \text{(Eq. 6)}$$

$$\dot{q}_k(t_f) = 0 \qquad \text{(Eq. 7)}$$

Cubic Polynomial Scheme (PTP) – *cont'd*

By combining the four constraints, Eqs. (4-7), with the Eqs. (1-2), the following equations for the coefficients result:

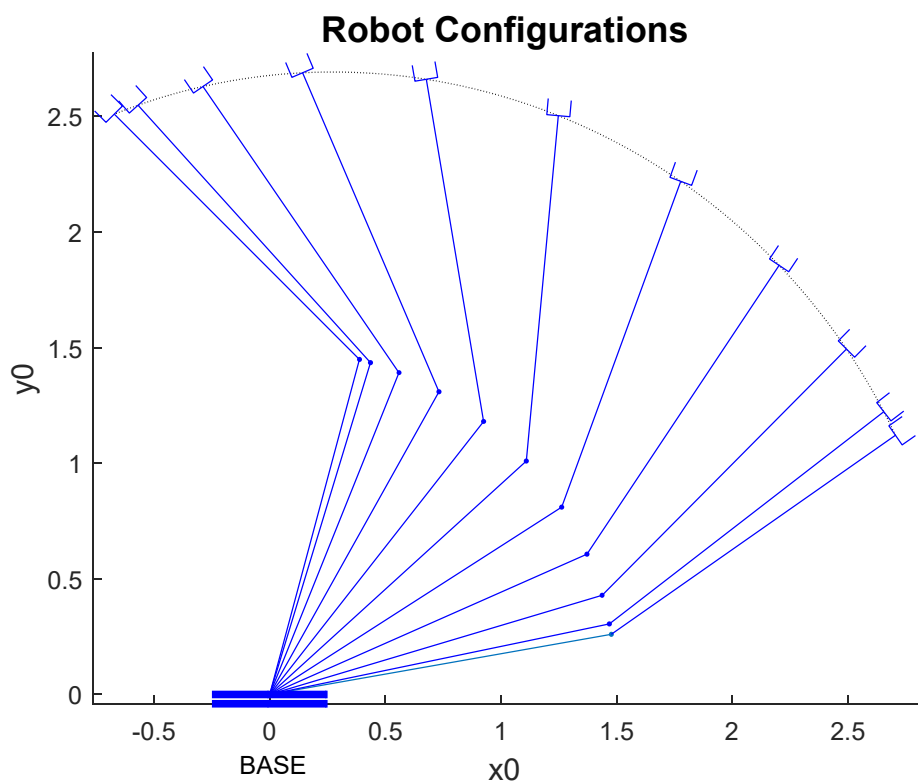
$$
\begin{aligned}
a_0 &= q_{k_0} \\
a_1 &= 0 \\
a_2 &= \frac{3}{{t_f}^2}\left(q_{k_f} - q_{k_0}\right) \\
a_3 &= -\frac{2}{{t_f}^3}\left(q_{k_f} - q_{k_0}\right)
\end{aligned}
\qquad \text{(Eq. 8)}
$$

For a generic case, $\dot{q}_k(t_0) = \dot{q}_{k_0}$ and $\dot{q}_k(t_f) = \dot{q}_{k_f}$ the coefficients are

$$
\begin{aligned}
a_0 &= q_{k_0} \\
a_1 &= \dot{q}_{k_0} \\
a_2 &= \frac{3}{{t_f}^2}\left(q_{k_f} - q_{k_0}\right) - \frac{2}{t_f}\dot{q}_{k_0} - \frac{1}{t_f}\dot{q}_{k_f} \\
a_3 &= -\frac{2}{{t_f}^3}\left(q_{k_f} - q_{k_0}\right) + \frac{1}{{t_f}^2}\left(\dot{q}_{k_f} + \dot{q}_{k_0}\right)
\end{aligned}
\qquad \text{(Eq. 9)}
$$

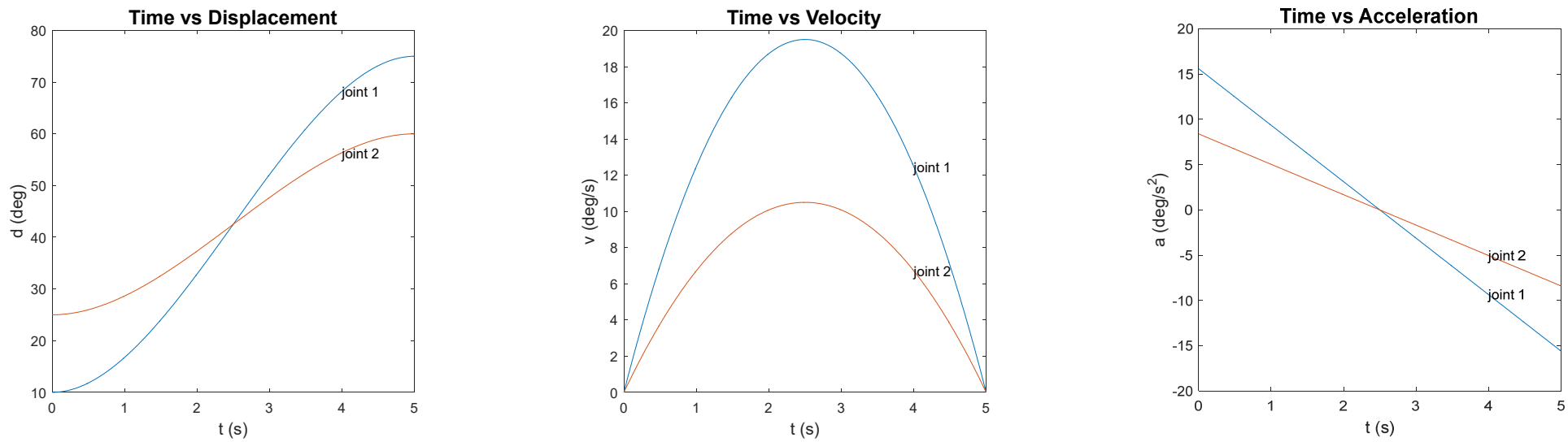
Cubic Polynomial Scheme (PTP)

Example: Assume a two-revolute jointed manipulator that starts from rest and stops at the end of the trajectory. The initial joint angles are $\theta_0 = [10 \quad 25]^\top$ and the final joint angles are $\theta_f = [75 \quad 60]^\top$, the duration of the trajectory is 5 sec. Determine the trajectory of the manipulator



Cubic Polynomial Scheme (PTP)

Example – cont'd: Assume a two-revolute jointed manipulator that starts from rest and stops at the end of the trajectory. The initial joint angles are $\theta_0 = [10 \quad 25]^\mathsf{T}$ and the final joint angles are $\theta_f = [75 \quad 60]^\mathsf{T}$, the duration of the trajectory is 5 sec.



Unfortunately, the acceleration profile is discontinuous at the beginning and end, joints would require infinite jerks.

Quintic Polynomial Scheme (PTP)

Higher polynomial schemes are often used in industrial applications.

$$q_k(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

$$\dot{q_k}(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4$$

$$\ddot{q_k}(t) = 2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3$$

This allows satisfying two more constraints: the initial and final conditions of the accelerations. Thus, the general coefficients are

$$a_0 = q_{k_0}$$

$$a_1 = \dot{q}_{k_0}$$

$$a_2 = \ddot{q}_{k_0}/2$$

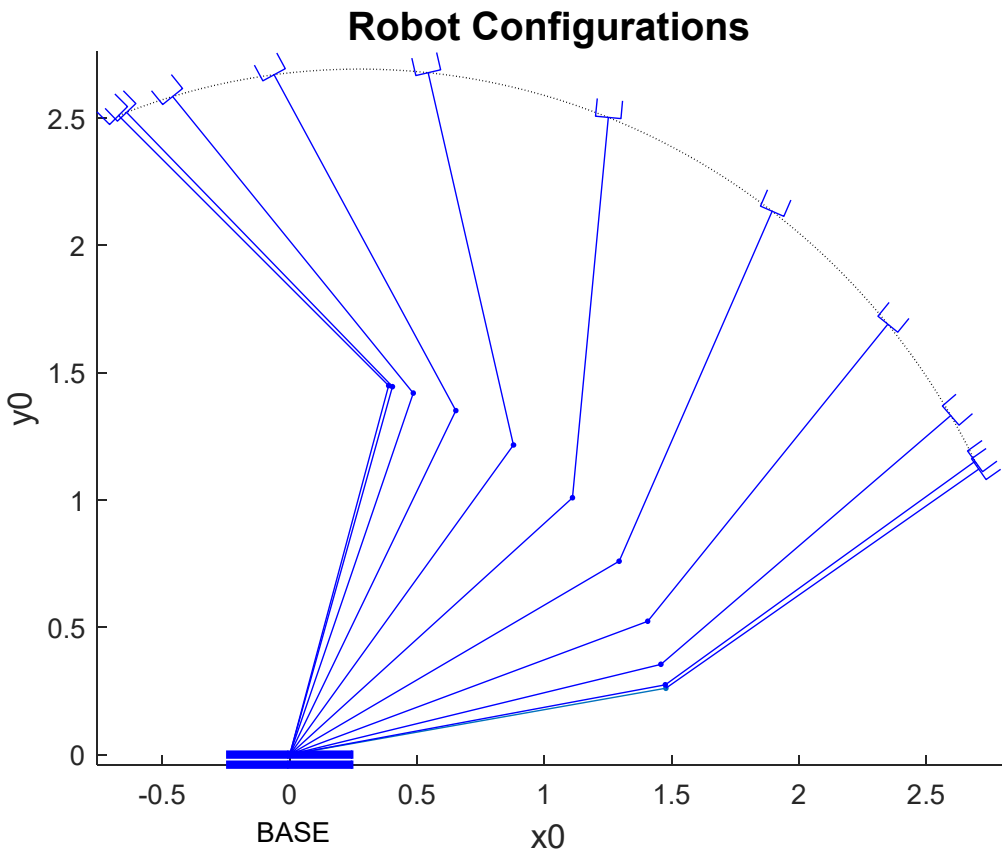
$$a_3 = \left(20\left(q_{k_f} - q_{k_0}\right) - \left(8\dot{q}_{k_f} + 12\dot{q}_{k_0}\right)t_f + \left(\ddot{q}_{k_f} - 3\ddot{q}_{k_0}\right){t_f}^2\right)/2{t_f}^3$$

$$a_4 = \left(-30\left(q_{k_f} - q_{k_0}\right) + \left(14\dot{q}_{k_f} + 16\dot{q}_{k_0}\right)t_f - \left(2\ddot{q}_{k_f} - 3\ddot{q}_{k_0}\right){t_f}^2\right)/2{t_f}^4$$

$$a_5 = \left(12\left(q_{k_f} - q_{k_0}\right) - \left(6\dot{q}_{k_f} + 6\dot{q}_{k_0}\right)t_f + \left(\ddot{q}_{k_f} - \ddot{q}_{k_0}\right){t_f}^2\right)/2{t_f}^5$$

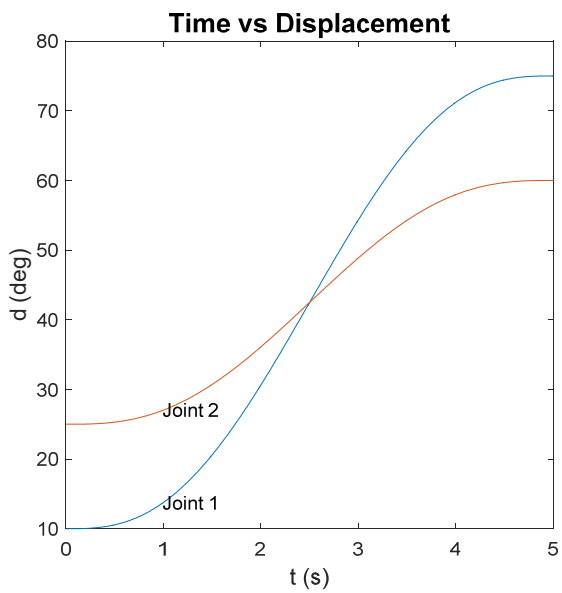
Quintic Polynomial Scheme (PTP)

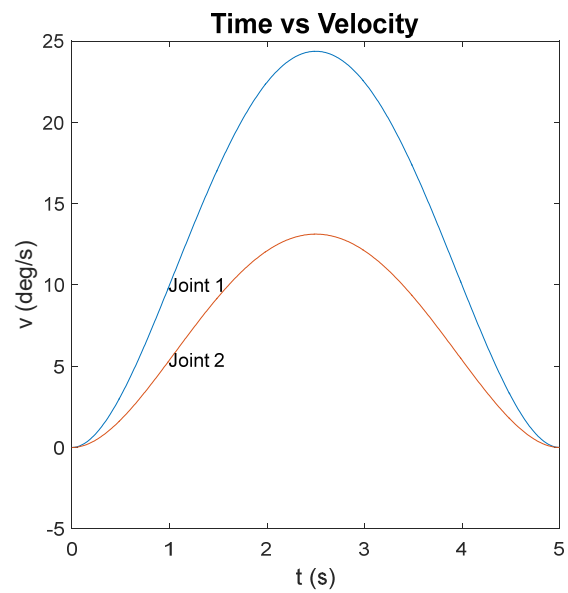
Example: Assume a two-revolute jointed manipulator that starts from rest and stops at the end of the trajectory. The initial joint angles are $\theta_0 = [10 \quad 25]^\top$ and the final joint angles are $\theta_f = [75 \quad 60]^\top$, the duration of the trajectory is 5 sec. Determine the trajectory of the manipulator using a quintic polynomial scheme.

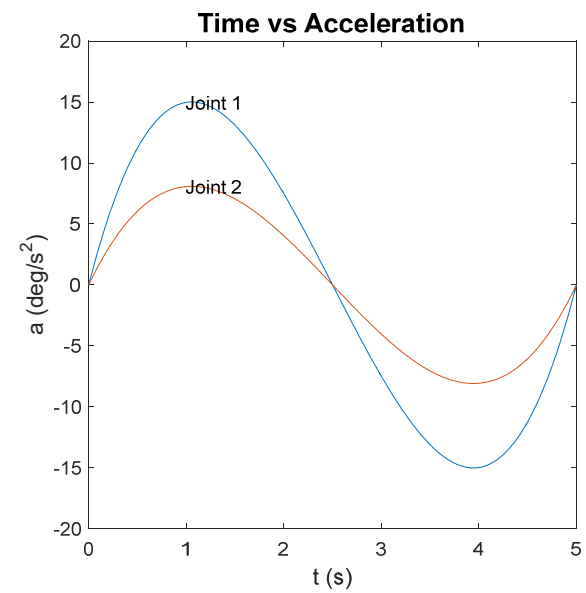


Quintic Polynomial Scheme (PTP)

Example: The displacement, velocity and acceleration profiles. Initial joint angles $\theta_0 = [10 \quad 25]^\top$ and final joint angles $\theta_f = [75 \quad 60]^\top$. Manipulator starts and ends at rest. Duration 5s.







Now both velocity and acceleration start and end at zero.

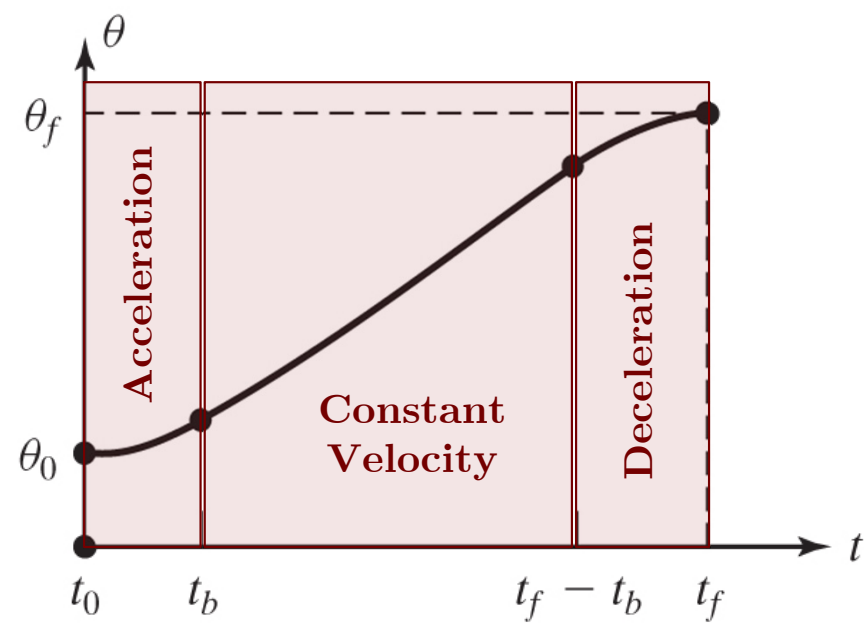
Trapezoidal Scheme (PTP)

This scheme is based on having a linear function with parabolic blends to maintain a smooth path, consequently the velocity profile will have a trapezoidal shape (linear-constant-linear) – Linear segment with parabolic blends.

The resulting trajectory is composed of three phases: acceleration in the initial phase, constant velocity in the middle phase and deceleration in the final phase.

In general, the duration of the acceleration and deceleration phases is equal, i.e., t_b.

<u>*Note*</u>: here the variables used are θ's but the same approach can be followed for prismatic joints, d's



Trapezoidal Scheme (PTP)

Since the duration of the acceleration and deceleration phases is equal, the magnitude of the acceleration in both phases is also equal, i.e.,

$$\left|\ddot{\theta}_k(t_b)\right| = \left|\ddot{\theta}_k\left(t_f - t_b\right)\right|$$

Similarly, the magnitude of the velocity during the transition of the phases must be the same, therefore

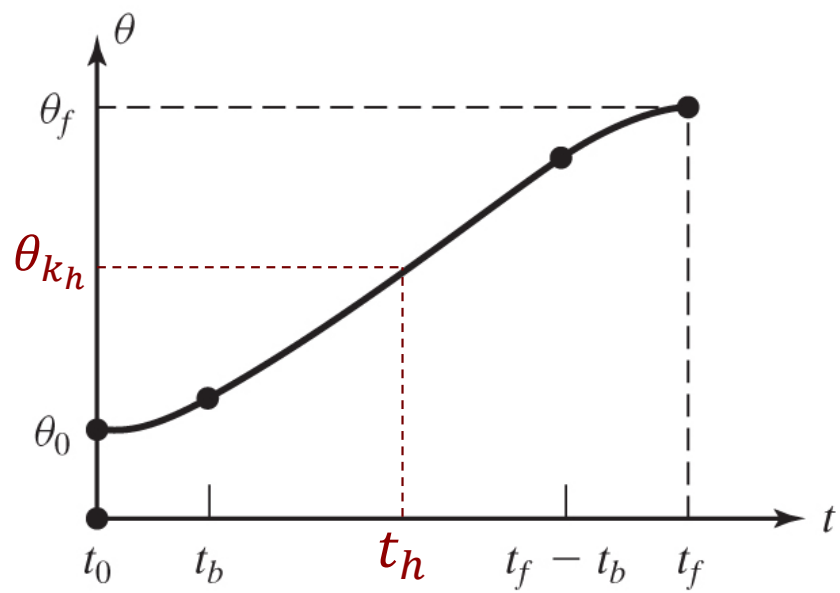
$$\ddot{\theta}_{k_b} t_b = \frac{\theta_{k_h} - \theta_{k_b}}{t_h - t_b}$$

where the subscript h represents the halfway point of the trajectory.

$$\theta_{k_h} = \frac{\theta_{k_0} + \theta_{k_f}}{2}$$

and,

$$t_h = \frac{t_f}{2}$$



Trapezoidal Scheme (PTP)

Since the blends are parabolic and $\dot{\theta}_k(t_0) = 0$, then θ_{k_b} is given by

$$\theta_{k_b} = \theta_{k_0} + \frac{1}{2}\ddot{\theta}_{k_b} {t_b}^2$$

Combining the previous two equations yields

$$\ddot{\theta}_{k_b} {t_b}^2 - \ddot{\theta}_{k_b} t_f t_b + \left(\theta_{k_f} - \theta_{k_0}\right) = 0 \qquad \text{(Eq. ※)}$$

where t_b and $\ddot{\theta}_{k_b}$ are unknown variables.

Usually, $\ddot{\theta}_{k_b}$ is prescribed and t_b is found with eq (Eq. ※), i.e.,

$$t_b = \frac{t_f}{2} - \frac{\sqrt{(\ddot{\theta}_{k_b} t_f)^2 - 4\ddot{\theta}_{k_b}\left(\theta_{k_f} - \theta_{k_0}\right)}}{2\ddot{\theta}_{k_b}}$$

Make sure the term inside the square root is positive.

Trapezoidal Scheme (PTP)

An approach to ensure the real solution of t_b, $\ddot{\theta}_{k_b}$ can be defined as

$$\ddot{\theta}_{k_b} = \frac{4\left(\theta_{k_f} - \theta_{k_0}\right)}{{t_f}^2} f_b \qquad \text{with} \qquad f_b \geq 1$$

Set f_b, find $\ddot{\theta}_{k_b}$, solve for t_b as follows,

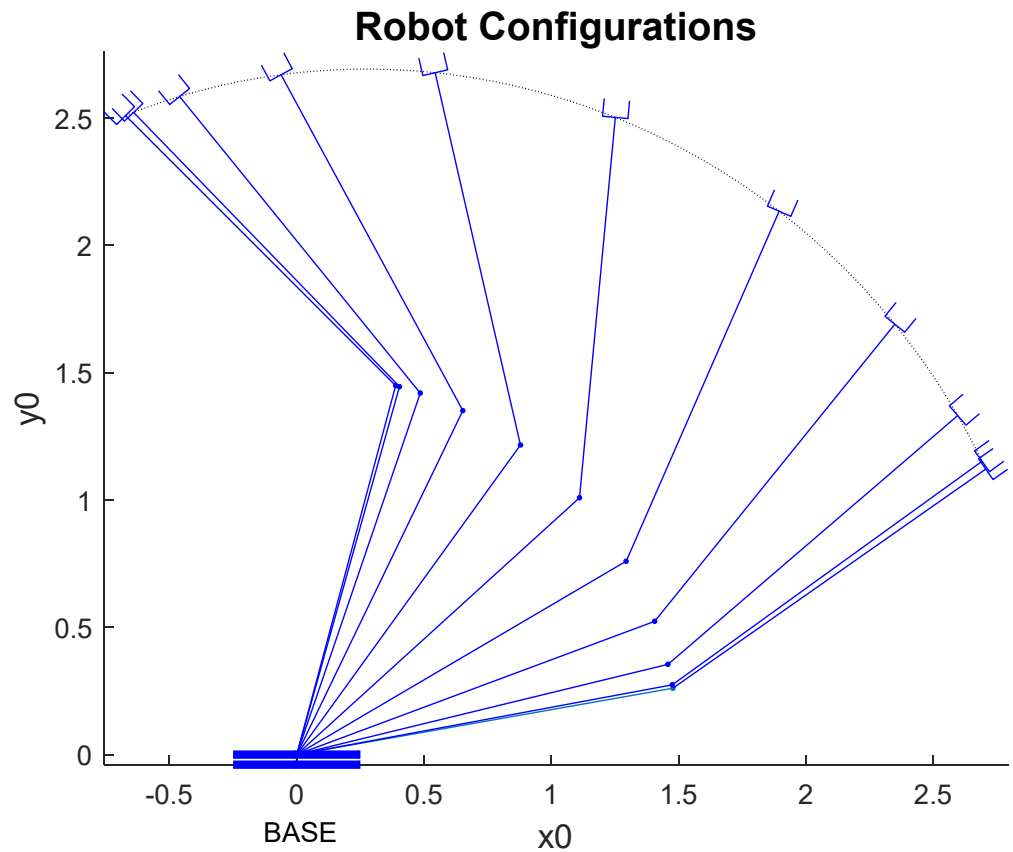
$$t_b = \min\left(roots[\ddot{\theta}_{k_b} \quad - \ddot{\theta}_{k_b} t_f \quad (\theta_{k_f} - \theta_{k_0})]\right)$$

Thus,

Phase	Displacement	Velocity	Acceleration
$0 \leq t \leq t_b$	$\theta_k(t) = \theta_{k_0} + \frac{1}{2}\ddot{\theta}_{k_b} t^2$	$\dot{\theta}_k(t) = \ddot{\theta}_{k_b} t$	$\ddot{\theta}_k(t) = \ddot{\theta}_{k_b}$
$t_b \leq t \leq t_f - t_b$	$\theta_k(t) = \theta_{k_0} + \ddot{\theta}_{k_b} t_b \left(t - \frac{t_b}{2}\right)$	$\dot{\theta}_k(t) = \ddot{\theta}_{k_b} t_b$	$\ddot{\theta}_k(t) = 0$
$t_f - t_b \leq t \leq t_f$	$\theta_k(t) = \theta_{k_f} - \frac{1}{2}\ddot{\theta}_{k_b}\left(t_f - t\right)^2$	$\dot{\theta}_k(t) = \ddot{\theta}_{k_b}\left(t_f - t\right)$	$\ddot{\theta}_k(t) = -\ddot{\theta}_{k_b}$

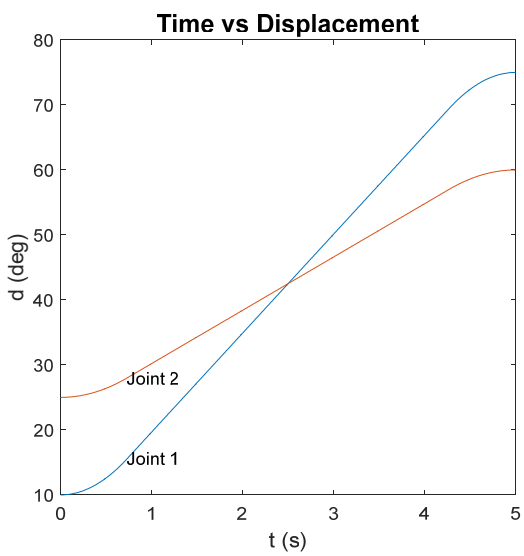
Trapezoidal Scheme (PTP)

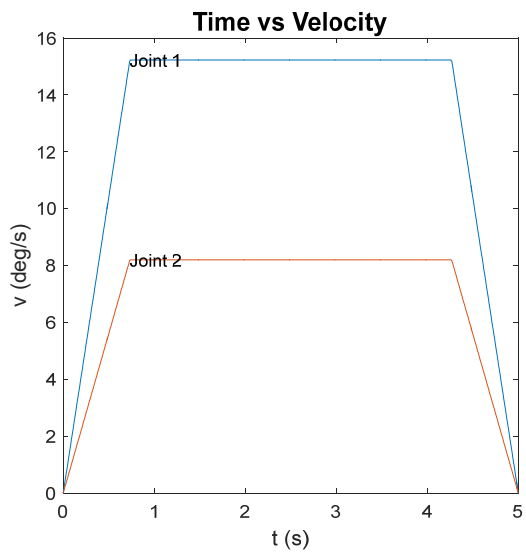
Example: Assume a two-revolute jointed manipulator that starts from rest and stops at the end of the trajectory. The initial joint angles are $\theta_0 = [10 \quad 25]^\top$ and the final joint angles are $\theta_f = [75 \quad 60]^\top$, the duration of the trajectory is 5 sec. Determine the trajectory of the manipulator using a trapezoidal scheme.

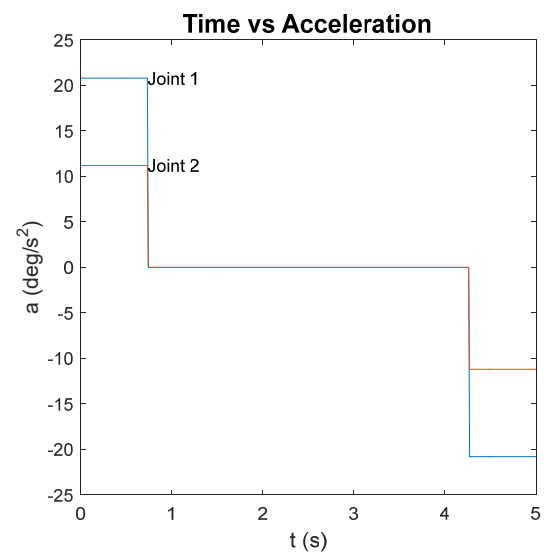


Trapezoidal Scheme (PTP)

Example: The displacement, velocity and acceleration profiles. Initial joint angles $\theta_0 = [10 \quad 25]^\top$ and final joint angles $\theta_f = [75 \quad 60]^\top$. Manipulator starts and ends at rest. Duration 5s.







Continuous Path (CP) Motion

There are applications in which the end-effector must follow a straight line, like in arc welding, spray painting, etc., in such cases continuous path motion is employed.

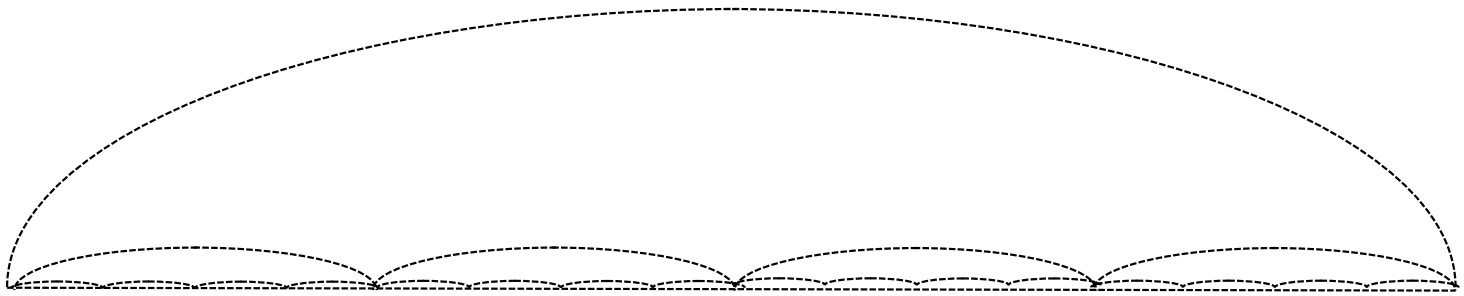
In continuous path motion (CP) motion, joints have to work together to accomplish the desired path.

A continuous path can be generated following two different methods

- **Via Points Scheme (Joint Space)**
- **Cartesian Scheme (Task Space)**

Via Points (CP)

A practical solution to generating a straight path is to divide the line into multiple points that are connected. Let the path followed by the end-effector be represented by the next animation:



This approach can be implemented with the schemes seen before. However, having individual segments implies that the manipulator stops at every point, which is not desired. In order to have a continuous motion without stopping, two approaches can be taken

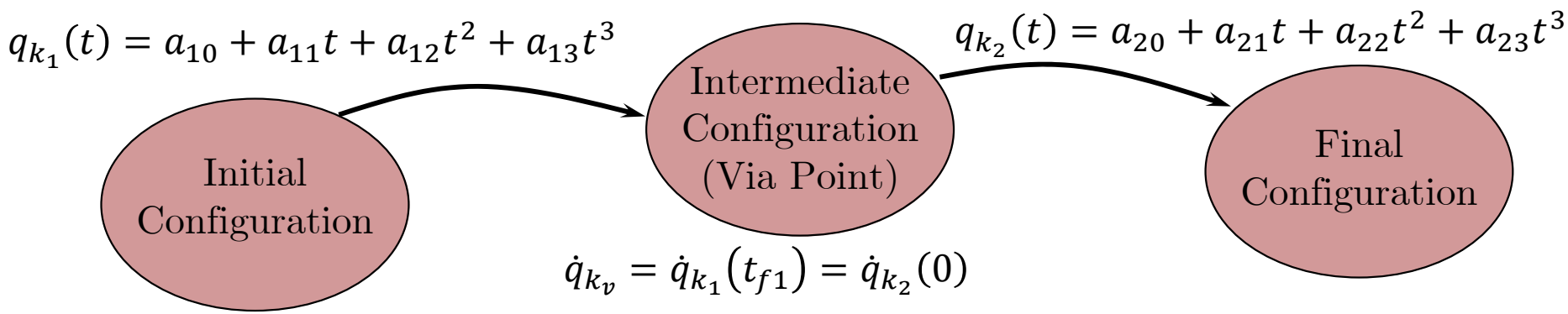
- Prescribed Velocity
- Matching Velocity and Accelerations

Via Points: Prescribed Velocity (CP)

Prescribed Velocity:

A simple solution is to prescribe the velocity between two joint segments, where a joint segment is defined by a polynomial scheme.

For example, assume that joint k starts from an initial configuration, passes through a via point, and ends its trajectory by reaching a final point. Each segment is represented by a different cubic scheme.



If the velocity at the via point is prescribed $\dot{q}_{k_v}$, then this velocity is the same as the velocity at the end of the first segment and at the beginning of the second segment. Each segment i duration is $0 \leq t \leq t_{f_i}$.

Via Points: Prescribed Velocity (CP)

Prescribed Velocity:

Polynomials

$$q_{k_1}(t) = a_{10} + a_{11}t + a_{12}t^2 + a_{13}t^3 \qquad q_{k_2}(t) = a_{20} + a_{21}t + a_{22}t^2 + a_{23}t^3$$

Conditions

$$q_{k_1}(0) = q_{k_0} \qquad q_{k_2}(0) = q_{k_v}$$

$$q_{k_1}(t_{f_1}) = q_{k_v} \qquad q_{k_2}(t_{f_2}) = q_{k_f}$$

$$\dot{q}_{k_1}(0) = 0 \qquad \dot{q}_{k_2}(0) = \dot{q}_{k_v}$$

$$\dot{q}_{k_1}(t_{f_1}) = \dot{q}_{k_v} \qquad \dot{q}_{k_2}(t_{f_2}) = 0$$

Coefficients

$$a_{10} = q_{k_0} \qquad a_{20} = q_{k_v}$$

$$a_{11} = 0 \qquad a_{21} = \dot{q}_{k_v}$$

$$a_{12} = \frac{3}{{t_{f_1}}^2}\left(q_{k_v} - q_{k_0}\right) - \frac{1}{t_{f_1}}\dot{q}_{k_v} \qquad a_{22} = \frac{3}{{t_{f_2}}^2}\left(q_{k_f} - q_{k_v}\right) - \frac{2}{t_{f_2}}\dot{q}_{k_v}$$

$$a_{13} = -\frac{2}{{t_{f_1}}^3}\left(q_{k_v} - q_{k_0}\right) + \frac{1}{{t_{f_1}}^2}\left(\dot{q}_{k_v}\right) \qquad a_{23} = -\frac{2}{{t_{f_2}}^3}\left(q_{k_f} - q_{k_v}\right) + \frac{1}{{t_{f_2}}^2}\left(\dot{q}_{k_v}\right)$$

Via Points: Prescribed Velocity (CP)

Example: Let $\theta_0 = 5^o$, $\theta_v = 15^o$, and $\theta_f = -10^o$. Assume the velocity at the via point is $\dot{\theta}_v = 0$. The duration per segment is 2 seconds.

Segment 1 $\dot{\theta}_0 = 0.$

$$a_{10} = \theta_0 = 5$$

$$a_{11} = \dot{\theta}_0 = 0$$

$$a_{12} = \frac{3}{t_{f_1}^2}(\theta_v - \theta_0) - \frac{2}{t_{f_1}}\dot{\theta}_0 - \frac{1}{t_{f_1}}\dot{\theta}_v$$

$$= \frac{3}{2^2}(15-5) = \frac{15}{2}$$

$$a_{13} = -\frac{2}{t_{f_1}^3}(\theta_v - \theta_0) - \frac{1}{t_{f_1}^2}\left(\dot{\theta}_0 + \dot{\theta}_v\right)$$

$$= -\frac{2}{2^3}(15-5) = -\frac{5}{2}$$

Segment 2 $\dot{\theta}_g = 0.$

$$a_{20} = \theta_v = 15$$

$$a_{21} = \dot{\theta}_v = 0$$

$$a_{22} = \frac{3}{t_{f_2}^2}(\theta_g - \theta_v) - \frac{2}{t_{f_2}}\dot{\theta}_v - \frac{1}{t_{f_2}}\dot{\theta}_g$$

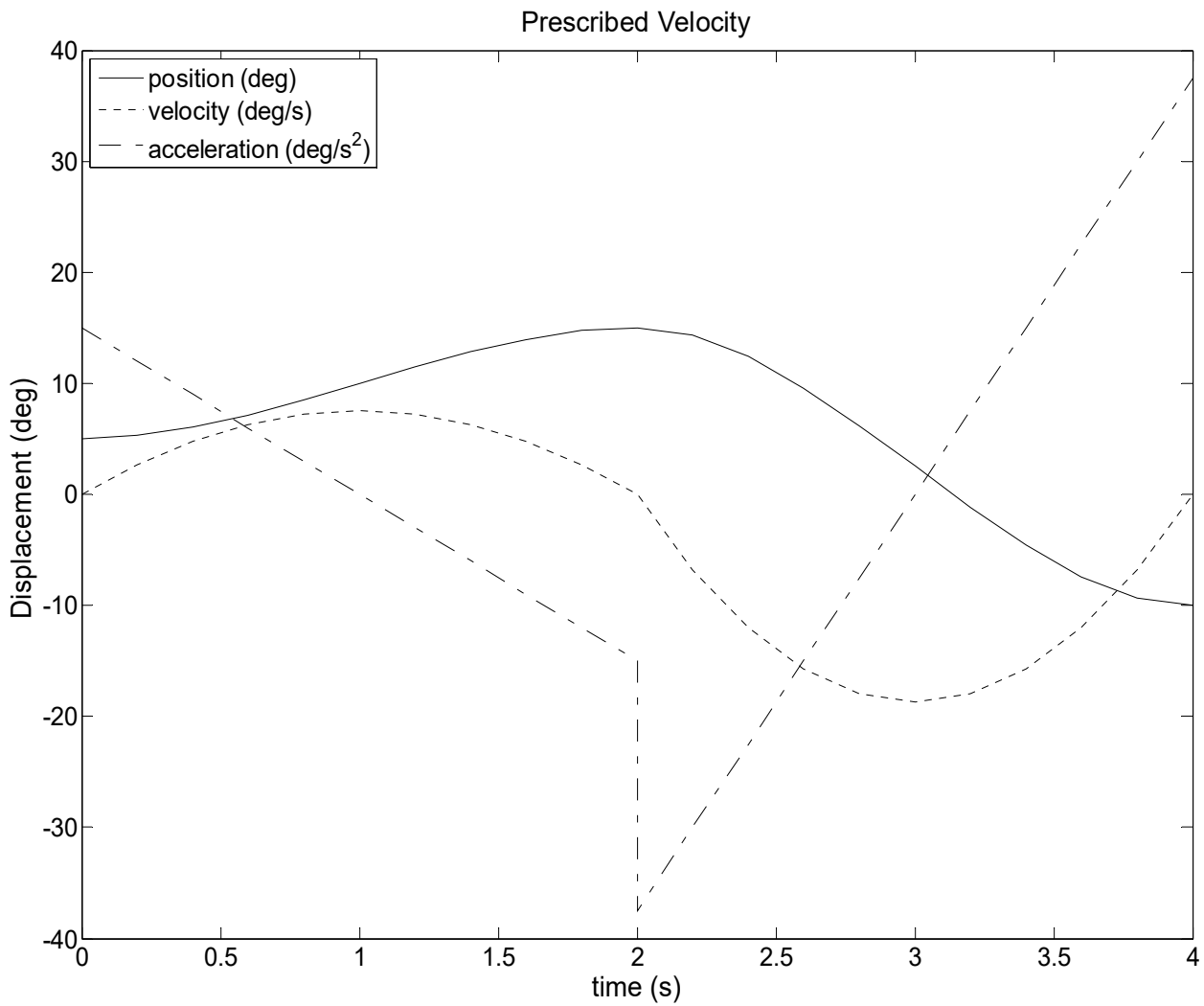
$$= \frac{3}{2^2}(-10-15) = -\frac{75}{4}$$

$$a_{23} = -\frac{2}{t_{f_2}^3}(\theta_g - \theta_v) - \frac{1}{t_{f_2}^2}\left(\dot{\theta}_g + \dot{\theta}_v\right)$$

$$= -\frac{2}{2^3}(-10-15) = \frac{25}{4}$$

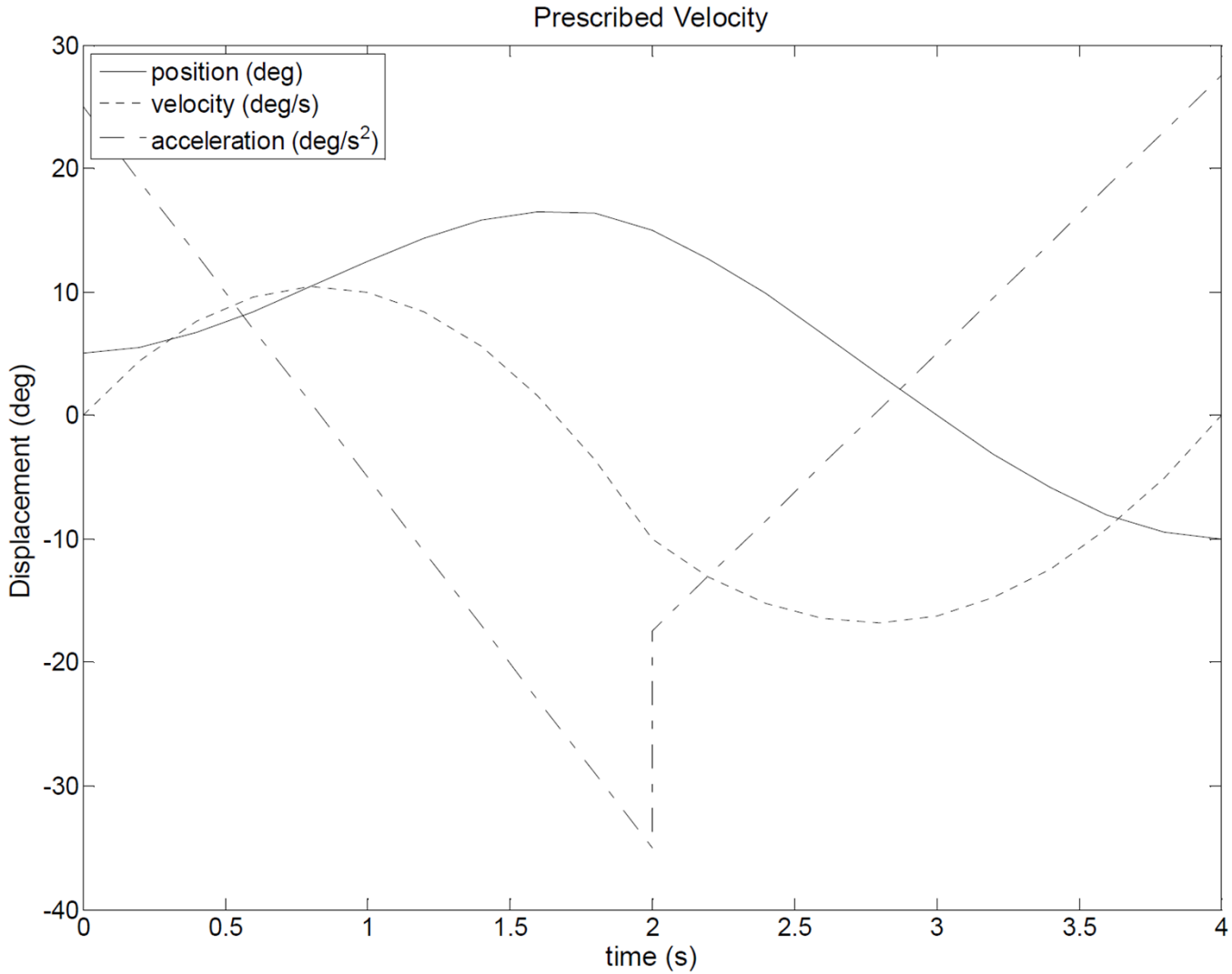
Via Points: Prescribed Velocity (CP)

Example – cont'd:



Via Points: Prescribed Velocity (CP)

Example – cont'd: For the same problem, now assume the velocity at the via point is $\dot{\theta}_v = -10$.



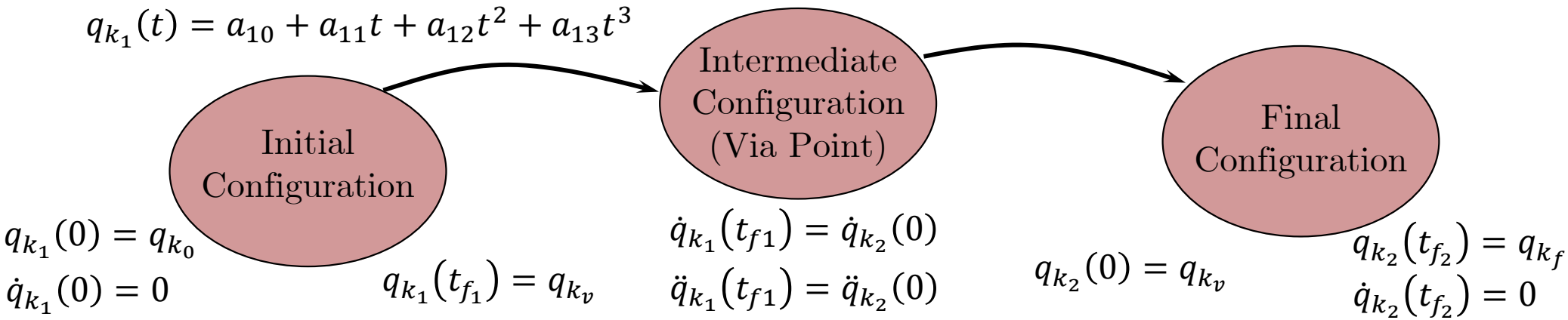
Matching Velocities and Accelerations

By prescribing the velocity at the via points and using cubic schemes the acceleration profile is discontinuous.

As an alternative, let us combine conditions between segments and define that the velocity and the accelerations between segments must be the same.

For example, for a two-cubic polynomial trajectory, there are 8 coefficients, therefore 8 conditions that can be satisfied:

$$q_{k_1}(t) = a_{10} + a_{11}t + a_{12}t^2 + a_{13}t^3$$



Matching Velocities and Accelerations – cont'd

Polynomials:

$$q_{k_1}(t) = a_{10} + a_{11}t + a_{12}t^2 + a_{13}t^3 \qquad q_{k_2}(t) = a_{20} + a_{21}t + a_{22}t^2 + a_{23}t^3$$

Conditions:

Segment I

$$\begin{cases} q_{k_1}(0) = q_{k_0} \\ q_{k_1}(t_{f_1}) = q_{k_v} \\ \dot{q}_{k_1}(0) = 0 \end{cases} \qquad \begin{aligned} &a_{10} = q_{k_0} \\ &a_{10} + a_{11}t_{f_1} + a_{12}{t_{f_1}}^2 + a_{13}{t_{f_1}}^3 = q_{k_v} \\ &a_{11} = 0 \end{aligned}$$

Segment II

$$\begin{cases} q_{k_2}(0) = q_{k_v} \\ q_{k_2}(t_{f_2}) = q_{k_f} \\ \dot{q}_{k_2}(t_{f_2}) = 0 \end{cases} \qquad \begin{aligned} &a_{20} = q_{k_v} \\ &a_{20} + a_{21}t_{f_2} + a_{22}{t_{f_2}}^2 + a_{23}{t_{f_2}}^3 = q_{k_f} \\ &a_{21} + 2a_{22}t_{f_2} + 3a_{23}{t_{f_2}}^2 = 0 \end{aligned}$$

Matching Vel/Acc

$$\begin{cases} \dot{q}_{k_1}(t_{f_1}) = \dot{q}_{k_2}(0) \\ \ddot{q}_{k_1}(t_{f1}) = \ddot{q}_{k_2}(0) \end{cases} \qquad \begin{aligned} &a_{11} + 2a_{12}t_{f_1} + 3a_{13}{t_{f_1}}^2 = a_{21} \\ &2a_{12} + 6a_{13}t_{f_1} = 2a_{22} \end{aligned}$$

Matching Velocities and Accelerations – cont'd

Consequently, all the equations must be solved simultaneously. This can be done in matrix form as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & t_{f_1} & {t_{f_1}}^2 & {t_{f_1}}^3 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & t_{f_2} & {t_{f_2}}^2 & {t_{f_2}}^3 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2t_{f_2} & 3{t_{f_2}}^2 \\ 0 & 1 & 2t_{f_1} & 3{t_{f_1}}^2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 6t_{f_1} & 0 & 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} a_{10} \\ a_{11} \\ a_{12} \\ a_{13} \\ a_{20} \\ a_{21} \\ a_{22} \\ a_{23} \end{bmatrix} = \begin{bmatrix} q_{k_0} \\ q_{k_v} \\ 0 \\ q_{k_v} \\ q_{k_f} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The solution of the system when $t_f = t_{f_1} = t_{f_2}$ is given in Craig's book.

$$a_{10} = \theta_0, \qquad a_{20} = \theta_\upsilon,$$

$$a_{11} = 0, \qquad a_{21} = \frac{3\theta_g - 3\theta_0}{4t_f},$$

$$a_{12} = \frac{12\theta_\upsilon - 3\theta_g - 9\theta_0}{4t_f^2}, \qquad a_{22} = \frac{-12\theta_\upsilon + 6\theta_g + 6\theta_0}{4t_f^2},$$

$$a_{13} = \frac{-8\theta_\upsilon + 3\theta_g + 5\theta_0}{4t_f^3}, \qquad a_{23} = \frac{8\theta_\upsilon - 5\theta_g - 3\theta_0}{4t_f^3}.$$

Via Points: Match Vel/Acc (CP)

Example: Let $\theta_0 = 5^o$, $\theta_v = 15^o$, and $\theta_f = -10^o$. Match velocities & accelerations at the via. The duration per segment is 2 seconds.

Segment 1 $\dot{\theta}_0 = 0.$

$$a_{10} = \theta_0 = 5$$

$$a_{11} = 0$$

$$a_{12} = \frac{12\theta_v - 3\theta_g - 9\theta_0}{4t_{f_1}^2}$$

$$= \frac{12(15) - 3(-10) - 9(5)}{4(2^2)} = 10.31$$

$$a_{13} = \frac{-8\theta_v + 3\theta_g + 5\theta_0}{4t_{f_1}^3}$$

$$= \frac{-8(15) + 3(-10) + 5(5)}{4(2^3)} = -3.91$$

Segment 2 $\dot{\theta}_g = 0.$

$$a_{20} = \theta_v = 15$$

$$a_{21} = \frac{3\theta_g - 3\theta_0}{4t_{f_2}}$$

$$= \frac{3(-10) - 3(5)}{4(2)} = -5.63$$

$$a_{22} = \frac{-12\theta_v + 6\theta_g + 6\theta_0}{4t_{f_2}^2}$$

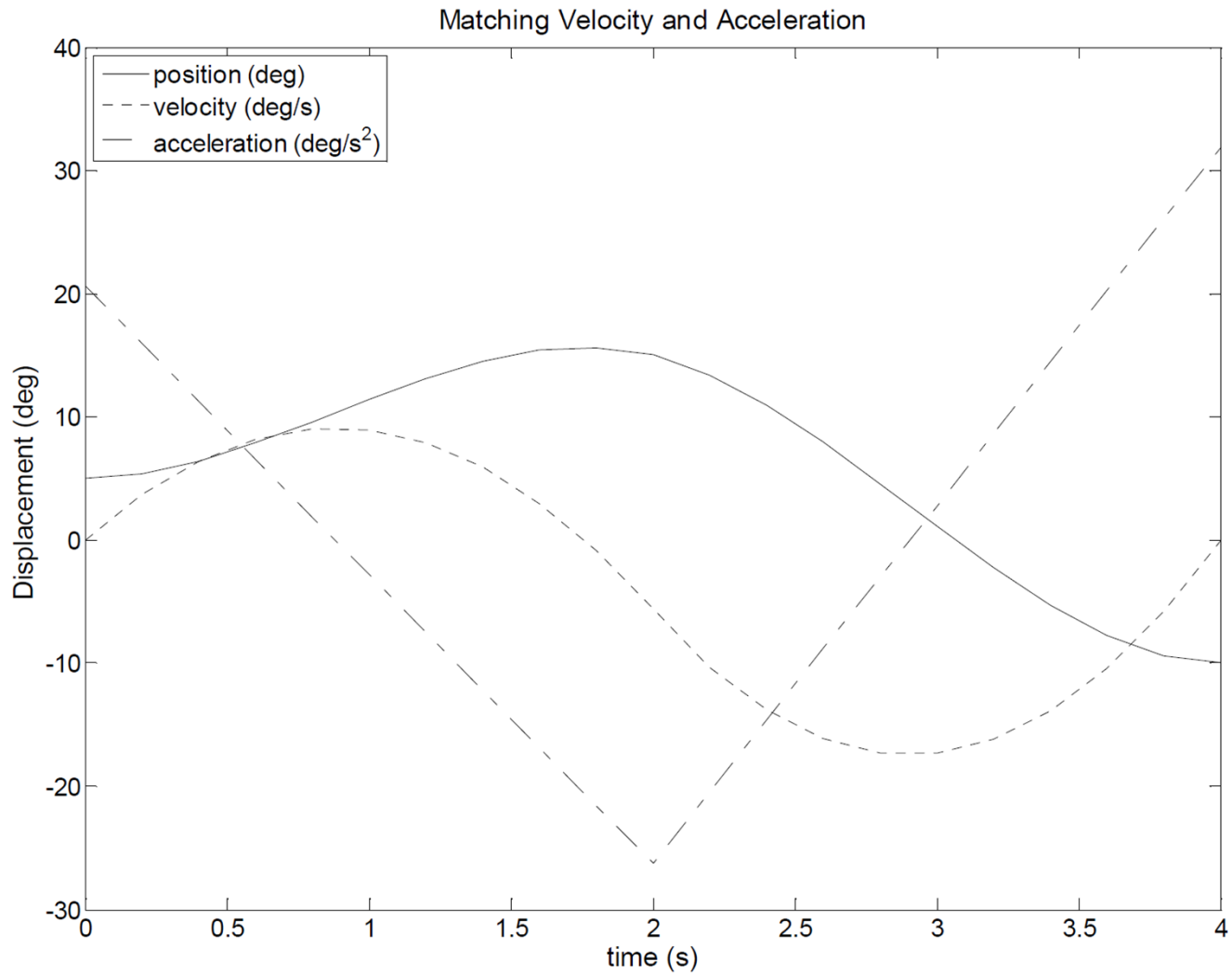
$$= \frac{-12(15) + 6(-10) + 6(5)}{4(2^2)} = -13.13$$

$$a_{23} = \frac{8\theta_v - 5\theta_g - 3\theta_0}{4t_{f_2}^3}$$

$$= \frac{8(15) - 5(-10) - 3(5)}{4(2^3)} = 4.84$$

Via Points: Match Vel/Acc (CP)

Example: Let $\theta_0 = 5^o$, $\theta_v = 15^o$, and $\theta_f = -10^o$. Match velocities & accelerations at the via. The duration per segment is 2 seconds.



Cartesian Scheme (CP)

Another way to establish the motion of a desired path of the end-effector is to generate the trajectory in the task space, ${}^{0}_{ee}\mathbf{T}_0 \rightarrow {}^{0}_{ee}\mathbf{T}_f$.

Note that one cannot generate a trajectory of a rotation matrix ${}^{0}_{ee}\mathbf{R}$ because by generating a trajectory for each element, nothing ensures that the resulting matrix maintains the properties of ${}^{0}_{ee}\mathbf{R}$.

Thus, the trajectory is generated with the independent elements of the position vector x, y, z and the Euler angles α, β, and γ, or the parameters of the equivalent angle/axis method or quaternions.

Since the joint variables are the input variables of a manipulator, then the inverse kinematics and the inverse velocity/acceleration problems must be solved at each moment

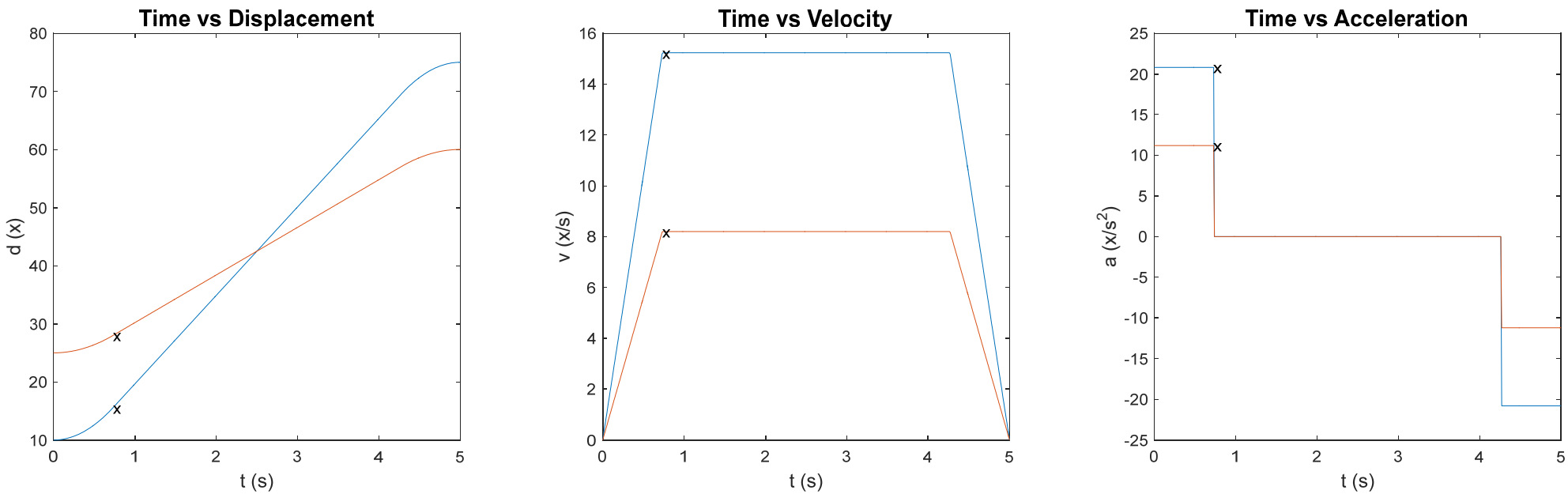
$$\dot{\boldsymbol{q}} = \mathbf{J}^{-1}\,\dot{\boldsymbol{x}}$$

This solution fails in case the manipulator is in a singular configuration

Cartesian Scheme (CP)

A common scheme adopted for in the Cartesian space is the Trapezoidal scheme.

The main advantage is that during the constant phase, the inverse velocity has to be solved only once. The linear part can be determined based on t_b.



Let The Learning Continue